Power

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Compare how quickly energy is transferred.

SUCCESS CRITERIA - I can...

1. Distinguish energy in J from power in W.
2. Use power=energy/time.
3. Explain equal work done at different rates.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “power” and “work” to explain your answer. Sentence starters are optional.

power: rate of energy transfer

work: energy transferred by a force along a displacement

interval: the time or range between two stated values

01 Warm-up: bring back an earlier idea

Supplied data: m=**0.3 kg**, drop h=**0.5 m**, final speed **1.5 m/s** from rest, g=**9.8**. **Find mgh**, translational K and the fraction appearing as translation.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: P_{avg} = W/Δt = energy transferred/Δt; P = Fv. Second equation: force parallel to velocity. Power units: **1 W** = **1 J/s**.

Worked reasoning: Two motors each transfer **120 J**. One takes **3 s**, the other **6 s**. Their average powers are **40** and **20 W**; total transferred energy is equal.

Your turn: Draw equal energy-transfer arrows and **label the different times** and powers.

Sentence starter: Both … . They differ in … because … .

Case	Energy (J)	Time (s)	Power (W)
1	____	____	____
2	____	____	____

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Two motors each transfer **120 J**. One takes **2 s**; the other **6 s**. Which has greater average power?

Sentence starter: Both … . They differ in … because … .

Motor	Energy	Time
A	120 J	2 s
B	120 J	6 s

Marzano check - circle: ① ② ③

04 Apply the physics

A **60 kg** climber gains **4 m** of vertical height in **8 s**.
1. Use g = **9.8 N/kg**.
2. **Calculate gravitational energy gain** and average mechanical power.
3. **Label the energy** and time in the table.

Sentence starter: I found … [unit]. This value means … in this situation.

Case	Energy (J)	Time (s)	Power (W)
1	____	____	____
2	____	____	____

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A **100 W** label is described as "**100 joules**." Correct the statement and **give time information** that would make an energy calculation possible.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A **4 kg** object slows from **8** to **6 m/s**. **Find net work** and **check its sign**.
Show the energy changes and transfers; **label units**.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

07 Exit check

A constant **200 N** forward force acts at steady **3 m/s**. **Find mechanical power** and work in **5 s**.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Case	Energy (J)	Time (s)	Power (W)
1	____	____	____
2	____	____	____

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A motor lifts a **20 kg** load **3 m** in **4 s** at constant speed; g=**10**. **Find useful power**. If efficiency is **75%**, find input power and energy not delivered to the load.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Case	Energy (J)	Time (s)	Power (W)
1	____	____	____
2	____	____	____

Marzano check - circle: ① ② ③